Vol 14 Chapter 11 — Information Completeness

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

Chapter 11 — Information Completeness

Level 1 — one sentence

BST asserts substrate information completeness: D_IV^5 + BST primaries $\{rank, N_c, n_C, C_2, g\}$ provide the COMPLETE information channel for observable physics — no hidden variables, no super-substrate, no missing degrees of freedom — falsifiable by identifying any observable BST cannot derive.

Level 2 — graduate-physicist precision

11.1 Statement of completeness

Substrate Information Completeness Hypothesis (Casey-named META-Hypothesis #1, 2026-05-24 — empirical version per Casey decision "use the strongest we can stand behind"; categorically separated from substrate-physics principles #1-#10 per Cal #120 framing recommendation):

> All physically observable quantities verified to date are derivable from D_IV^5 structure and the five BST primary integers ($rank = 2, N_c = 3, n_C = 5, C_2 = 6, g = 7$), with no additional information required. As of 2026-05-24: 600+ predictions verified within stated precision, zero clean falsifier found, one open candidate-falsifier (baryogenesis η_B).

Categorical note (Cal #120): This is a META-Hypothesis — a claim the framework makes about ITSELF (expressive completeness/closure), categorically different from substrate-physics principles #1-#10 (which describe what the substrate IS/DOES). Numbered in a separate META-N slot to preserve Cal #99 META-theorem discipline.

Casey directive 2026-05-24 — PROOF ATTEMPT v0.1 IN FLIGHT (Lyra task #321, filed Sunday afternoon):

Theorem InfoCompleteness-1 (v0.1 Route A): Let $\mathcal{A}_{sub}$ be the substrate operator algebra on Bergman $H^2(D_{IV}^5)$ generated by the 14-operator substrate zoo (position + momentum + spin + angular momentum + Hamiltonian + Bell-CHSH + T + C + P + N + γ^5 + Q + C_3 + I_3) + 5 BST primary integers + N_max=137. Then every observable in the Standard Model + every BST-derivable observable across all scales (electron, nuclear, atomic, molecular, biological, cosmological) is expressible as element of $\mathcal{A}_{sub}$.

Equivalent: D_IV^5 + 5 primaries are INFORMATIONALLY COMPLETE for observable physics — no additional structure required beyond substrate specification + operator algebra.

Proof skeleton (4 lemmas, Lyra v0.1): - Lemma A.1 Finite generating set: $\mathcal{A}_{sub}$ generated by 14 substrate-native operators (T2419 + T2421 + T2422 + T2425 + T2433 + T2434 + T2470 + T2471 + T2472 + T2399 + Elie K52a H_sub + 3 candidate operators) - Lemma A.2 Generating set sufficiency: every SM operator expressible as polynomial in 14 generators via Lie-algebra closure (multi-month formal verification pending) - Lemma A.3 Cross-scale extension: atomic + molecular + biological + cosmological observables reduce to $\mathcal{A}_{sub}$ elements per Cross-Scale Invariance Investigation v0.4 - Lemma A.4 Closure: A.1 + A.2 + A.3 → D_IV^5 + 5 primaries informationally complete → QED

Two routes: - Route A (representation-theoretic, Lyra preferred): 14-operator zoo + Lie-algebra spanning. Depends on SP-31-1 (Hilbert space spec) + SP-31-6 (operator zoo completion). - Route B (empirical + decidability, fallback): formal enumeration of "all observables" + decidability + per-observable D_IV^5-derivability. We already have this empirically (600+ predictions).

Critical test: η_B baryogenesis is the open observable. If Lyra's framework derives η_B from $\mathcal{A}_{sub}$ → strong confirmation; if not derivable in $\mathcal{A}_{sub}$ → falsifies completeness.

Status: I-tier v0.1 proof framework filed; multi-month formal closure (~30% probability clean Route-A proof in 6-12 months). D-tier promotion gated on Lemma A.2 rigorous verification + η_B closure.

11.2 Free parameter count

- Standard Model: 19+ free parameters (Yukawa couplings, mixing angles, neutrino masses, QCD theta, cosmological parameters)
- GR + Cosmology: 6+ free parameters (Λ, dark matter, baryon density, etc.)
- BST: 0 free parameters; everything derives from {rank, N_c, n_C, C_2, g}

11.3 Empirical support

Current BST status (as of 2026-05-24): - 600+ predictions verified within precision (Vol 0 Status) - All 7 Millennium problems addressed (RH conditional, others closed) - 100+ derived constants with explicit BST formulas - Substrate Strong-Uniqueness Theorem 11 RIGOROUSLY CLOSED + 7 candidates

11.4 Falsification

To falsify substrate completeness: identify an observable physical quantity that BST cannot derive.

Categories of candidate falsifiers (none yet successful): - New particle outside BST-predicted spectrum (e.g., SUSY, GUT proton decay, sterile neutrinos) - Constant cleanly outside BST-primary expressions (e.g., baryogenesis η_B — open in BST) - Tsirelson saturation (Bell at $2\sqrt{2}$ exact, contradicting BST 1/8 sub-Tsirelson)

Falsifiers BST predicts to be NULL (Five-Absence Predictions Set, Casey-named principle #2, canonical 5-item ordering per Casey Option 1 Friday 2026-05-22): GUT, proton decay, monopoles, sterile neutrinos, SUSY.

Note on DM: BST DERIVES dark matter as the Wallach shadow (16/3 at 0.2% precision), so DM is NOT in the absence set — BST has a positive DM prediction. The earlier K65 "4+1 scope" enumeration that included a sixth absence ("no NEW DM particle beyond Wallach shadow") has been retired in favor of the cleaner 5-item canonical per Casey 2026-05-22 decision.

11.5 Reduction-mathematics view (SP-20-3)

Reverse mathematics question: what minimum substrate is required to derive each physics theorem?

BST conjecture: D_IV^5 + 5 integers is the minimum sub-structure.

Open in BST: explicit per-theorem minimum-substrate enumeration (SP-20-3 BST-RM program).

11.6 K-audit anchors

- Strong-Uniqueness Theorem v0.10.5
- Five-Absence Predictions Set (Casey-named #2)
- SP-20 Information Completeness Program

Level 3 — 5th-grader accessibility

Information completeness: BST claims D_IV^5 + 5 integers $(rank, N_c, n_C, C_2, g) = (2, 3, 5, 6, 7)$ is the COMPLETE specification of observable physics. Standard Model has 19+ free parameters; BST has 0. Falsifier: find any observable BST can't derive. None yet. BST predicts NULL: GUT, proton decay, DM particle, monopoles, sterile neutrinos, SUSY (Five-Absence). Reverse math (SP-20-3): what minimum substrate suffices? Conjecture: D_IV^5 + 5 integers.

What comes next

Chapter 12 develops substrate-CI architecture.

Where to look this up

- BST Strong-Uniqueness Theorem
- Five-Absence Predictions Set
- SP-20 program